Introduction

Moments are the descending sequence of expectations $E[X], E[X^2], E[X^3], \ldots$ that summarise a distribution's location, spread, skewness, kurtosis, and finer shape features. The moment generating function (MGF) packages all moments into a single function; when it exists, the MGF characterises the distribution uniquely. MGFs give elegant proofs of the CLT, the distribution of sums of independent variables, and the closure properties of many distributional families.

Prerequisites

The reader should know what an expectation is and be comfortable with basic calculus.

Theory

The **k-th moment** of X is $\mu_k = E[X^k]$ (provided the integral converges). The **central moments** are $\mu_k^{(c)} = E[(X-\mu)^k]$; $\mu_2^{(c)}$ is the variance, $\mu_3^{(c)}$ feeds skewness, $\mu_4^{(c)}$ feeds kurtosis.

The **moment generating function** is

$$M_X(t) = E[e^{tX}], \qquad \text{for all } t \text{ where the expectation is finite.}$$

If M_X is finite in a neighbourhood of zero, then (i) all moments of X exist, (ii) the k-th moment equals the k-th derivative of M_X at zero, $M_X^{(k)}(0) = \mu_k$, and (iii) M_X determines the distribution uniquely.

Properties.

- For independent X, Y: $M_{X+Y}(t) = M_X(t)M_Y(t)$. MGFs turn sums into products.
- For affine transformations: $M_{aX+b}(t) = e^{bt}M_X(at)$.
- The MGF of a standard normal is $M(t) = e^{t^2/2}$. For the normal $\mathcal{N}(\mu, \sigma^2)$: $M(t) = e^{\mu t + \sigma^2 t^2/2}$.

Limitations. The MGF may not exist for all t – Cauchy's MGF is infinite everywhere except at $t = 0$. For such distributions, the **characteristic function** $\varphi_X(t) = E[e^{itX}]$ is preferred: it always exists and shares most useful properties with the MGF.

MGFs prove the CLT elegantly: the MGF of the standardised mean converges to $e^{t^2/2}$, which is the MGF of the standard normal, and uniqueness of MGFs gives convergence in distribution.

Assumptions

Existence of moments up to order k requires $E[|X|^k] < \infty$. Existence of the MGF in a neighbourhood of zero is a much stronger condition (light-tailed distributions only).

R Implementation

```r
set.seed(2026)
n <- 1e6

# Moments of a sample from a gamma distribution
x <- rgamma(n, shape = 3, rate = 2)
moments <- sapply(1:4, function(k) mean(x^k))
names(moments) <- paste0("m_", 1:4)
moments

# Theoretical moments: E[X^k] = Gamma(shape + k)/Gamma(shape) * rate^-k
shape <- 3; rate <- 2
theoretical <- sapply(1:4, function(k) {
```

```r
  gamma(shape + k) / gamma(shape) * rate^(-k)
})
names(theoretical) <- paste0("m_", 1:4)
theoretical

# MGF of gamma at a few t: M(t) = (1 - t/rate)^-shape for t < rate
t_vals <- c(0.1, 0.5, 1.0)
mgf_theoretical <- (1 - t_vals / rate)^(-shape)
mgf_empirical   <- sapply(t_vals, function(t) mean(exp(t * x)))
cbind(t = t_vals, theoretical = mgf_theoretical,
      empirical = mgf_empirical)
```

We compare empirical and theoretical moments and MGF values for a gamma distribution.

Output & Results

```
       m_1       m_2       m_3       m_4
   1.5001    3.0018    8.2566   32.9916

       m_1       m_2       m_3       m_4
   1.5000    3.0000    8.2500   33.0000

     t  theoretical  empirical
[1,] 0.1    1.2167    1.2168
[2,] 0.5    2.3704    2.3707
[3,] 1.0    8.0000    8.0076
```

Empirical and theoretical values agree to three decimal places with one million observations; with smaller samples the agreement is poorer in the higher moments and far tails.

Interpretation

For applied statistics, MGFs appear most often as proof tools rather than quantities to compute. Many closure properties (the normal family is closed under convolution; the gamma family has the same closure with a fixed rate parameter) are memorised via their MGFs.

Practical Tips

- For heavy-tailed distributions, higher moments may not exist; always check that the moments you are computing are finite.
- Characteristic functions always exist; use them when MGFs don't.
- In applied work, the first four moments (mean, variance, skewness, kurtosis) typically suffice; higher moments are noisy and rarely informative.
- The MGF of a sum of independent variables is the product of individual MGFs – a fact that makes many distributional derivations tractable.
- Do not confuse moments with cumulants: the cumulant generating function (log of the MGF) has additive behaviour under sums and is often preferred in theoretical work.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**

- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr